

C

LV.0 → LV.1 · AI ENGINEER

The 7 C's that move you from **Level 0** → **1** AI engineer.

CNN, Chunking, Chain of Thought and more. **Plain-English** definitions, one swipe each, plus the tools and channels that build your C-stack.

COST FUNCTION • FOUNDATIONS

Cost Function

predict • measure • minimize

A cost function is a mathematical function that measures **how far a model's predictions are from the actual values**. During training, optimization algorithms work to minimize this cost, enabling the model to improve its accuracy and learn the underlying patterns in the data.

USED IN

- Model training
- Optimization
- Gradient descent

model:

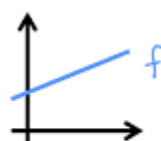
$$f_{w,b}(x) = wx + b$$

parameters:

$$w, b$$

cost function:

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$



PREDICTIONS VS ACTUAL

Clustering

group • by • similarity

Clustering algorithms are unsupervised methods that automatically **group data points by similarity** without any labels. By discovering natural structure in data, they power customer segmentation, anomaly detection, and exploratory analysis, with K-Means, DBSCAN, and hierarchical clustering among the most widely used.

USED IN

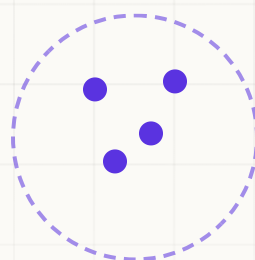
• Segmentation

• Anomaly detection

• K-Means



$k = 3$



UNSUPERVISED GROUPING

CONVOLUTIONAL NEURAL NETWORK • VISION

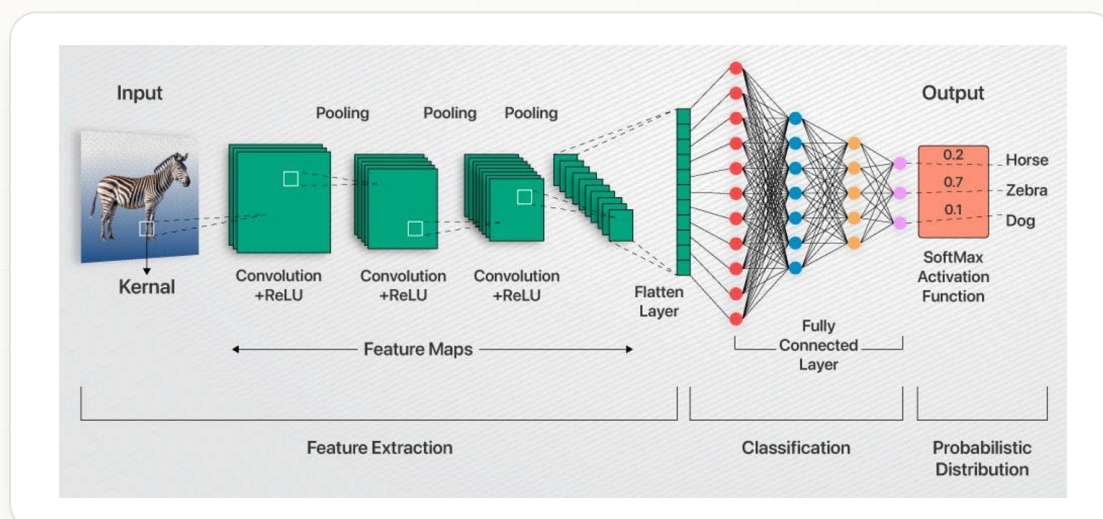
CNN

convolve • pool • classify

A Convolutional Neural Network is a deep learning architecture that **automatically detects patterns and features in visual data** through convolutional layers. Widely used in computer vision, CNNs power image classification, object detection, facial recognition, medical imaging, and autonomous driving.

USED IN

- Image classification
- Object detection
- Medical imaging



FEATURE EXTRACTION → CLASSIFY

CHAIN OF THOUGHT · COT · PROMPTING

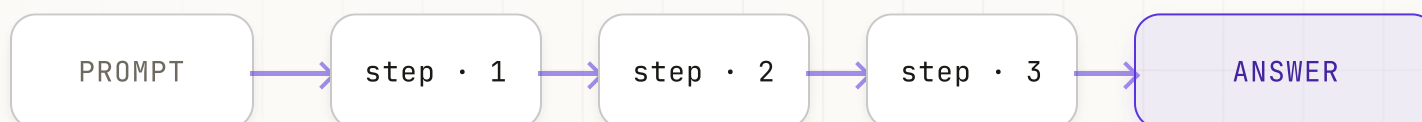
Chain of Thought

think · step · by · step

Chain of Thought is a prompting technique that encourages AI models to **reason through problems step by step** before giving a final answer. By breaking complex tasks into intermediate reasoning steps, it improves accuracy, logical consistency, and performance on math, coding, planning, and multi-step reasoning.

USED IN

- Math & code
- Planning
- Reasoning models



STEP-BY-STEP REASONING

Context Window

tokens • in • memory

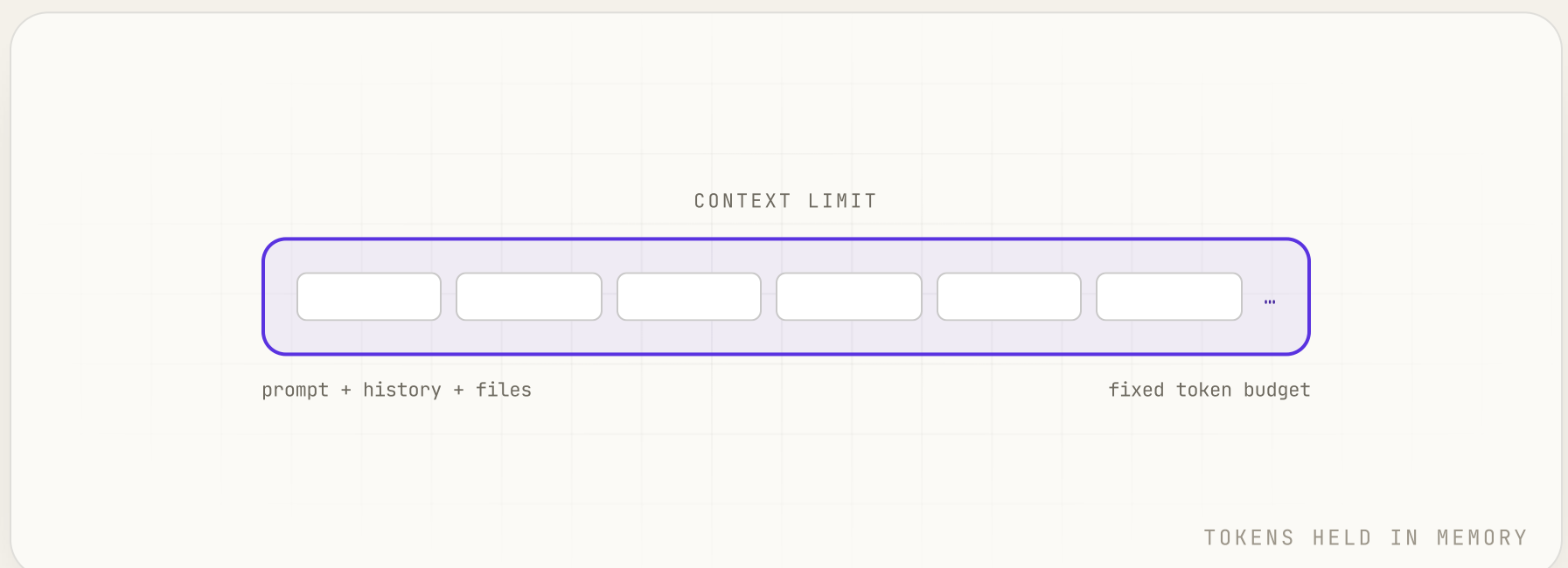
A context window is the **maximum amount of information a model can process in a single interaction**, including the prompt, conversation history, and uploaded content. Larger windows let models handle longer documents, stay coherent across extended chats, and reason over more complex tasks.

USED IN

• Long documents

• Long chats

• RAG



DOCUMENT CHUNKING • RETRIEVAL

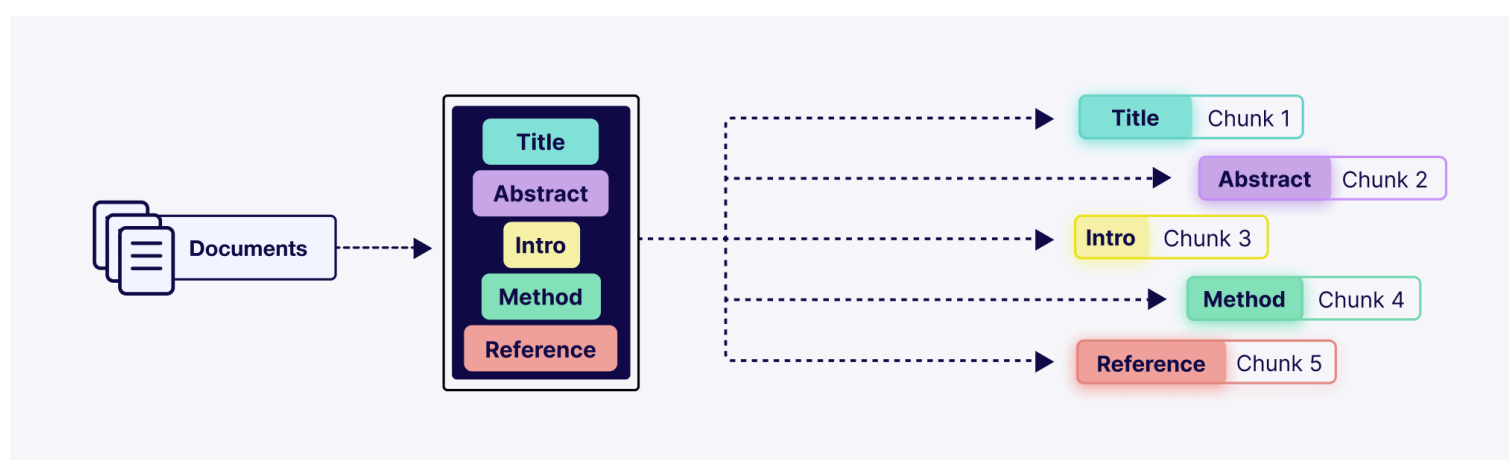
Chunking

split • embed • retrieve

Chunking is the process of **dividing large documents into smaller, manageable pieces** while preserving meaningful context. Widely used in Retrieval-Augmented Generation systems, it improves information retrieval, embedding quality, and AI responses by ensuring relevant content fits within a model's context window.

USED IN

- RAG
- Vector search
- Embeddings



DOCUMENTS → CHUNKS

COMPUTE UNIFIED DEVICE ARCHITECTURE • INFRASTRUCTURE

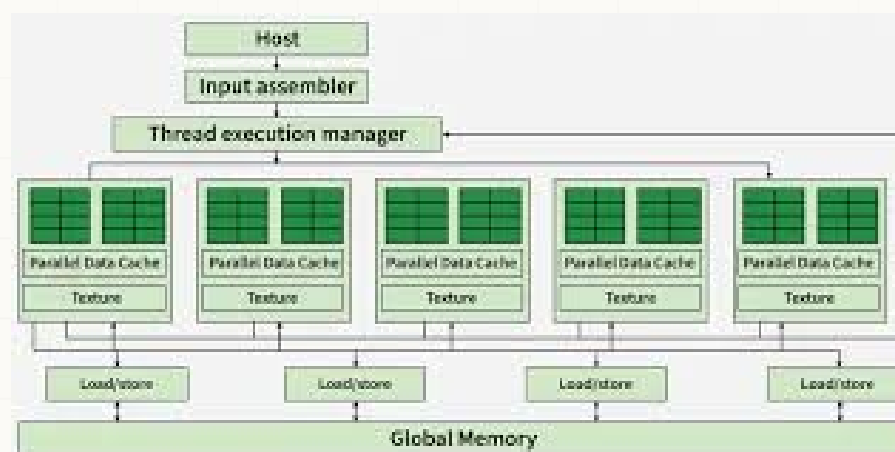
CUDA

parallel • GPU • compute

CUDA is NVIDIA's parallel computing platform and programming model that lets developers **harness GPU power for high-performance computing**. Widely used in AI and deep learning, it accelerates model training, inference, scientific simulations, and data processing by running thousands of computations simultaneously.

USED IN

- GPU training
- Deep learning
- High-perf computing



MASSIVELY PARALLEL

↓ UP NEXT • TOOLS

Tools next.

Beyond the 7 C-words sit the tools that **write your code, store your data, and run your agents** in production.

C

AI CODING ASSISTANT

Claude Code

CLAUDE
CODE

Anthropic • terminal-first dev

Claude Code is Anthropic's **AI-powered coding assistant built for terminal-first development**.

It helps developers write, edit, debug, and understand code, run project-wide tasks, and automate workflows through natural language, enabling faster, more efficient engineering across large codebases.

claude.com/code

[in /company/anthropicresearch](https://company/anthropicresearch)

WHAT IT IS

Anthropic's terminal-native coding agent.

IN AI STACKS

Pairs with your shell, git, and editor.

WHY DEVS USE IT

Edit, debug, and ship across whole codebases.

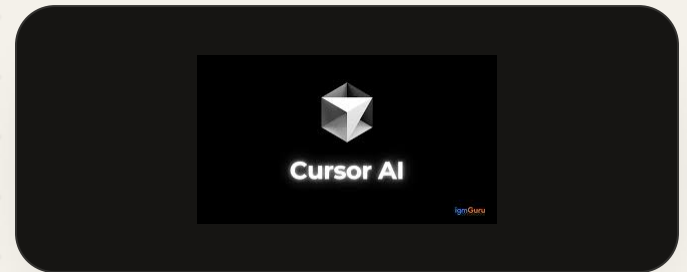
BEST FOR

Large refactors and project-wide tasks.

AI CODE EDITOR

Cursor

the AI-first IDE



Cursor is an AI-first code editor that builds large language models directly into the workflow. It **predicts edits, answers questions about your codebase, and writes code from natural-language prompts**, helping developers move faster while keeping full control of every change they ship.

[↗ cursor.com](https://cursor.com)

WHAT IT IS

An editor with an LLM built in.

IN AI STACKS

A daily driver for AI-assisted coding.

WHY DEVS USE IT

Predictive edits and codebase-aware chat.

BEST FOR

Fast iteration with full human control.

MULTI-AGENT FRAMEWORK

CrewAI

orchestrate AI agents



CrewAI is an open-source framework for orchestrating multiple AI agents that **collaborate to solve complex tasks**. By assigning specialized roles, goals, and workflows to autonomous agents, it lets developers build scalable, collaborative AI systems for research, automation, coding, and business processes.

➔ crewai.com

in [/company/crewai-inc](https://company/crewai-inc)

WHAT IT IS

An open-source multi-agent orchestrator.

IN AI STACKS

Coordinates research, coding, and automation.

WHY DEVS USE IT

Give agents roles, goals, and workflows.

BEST FOR

Collaborative, multi-step agent systems.

ENTERPRISE LLM PLATFORM

Cohere



LLMs and embeddings for business

Cohere is an enterprise AI company building large language models, embeddings, and retrieval systems for **secure, business-critical applications**. Its privacy-first approach, multilingual capabilities, and RAG-focused models help organizations build intelligent search, assistants, and knowledge systems at scale across industries.

[➔ cohere.com](https://cohere.com)

[in /company/cohere-ai](https://cohere.com/company/cohere-ai)

WHAT IT IS

Enterprise LLMs, embeddings, and retrieval.

IN AI STACKS

Powers search, assistants, and knowledge tools.

WHY DEVS USE IT

Privacy-first, multilingual, RAG-ready models.

BEST FOR

Secure, business-critical applications.

DISTRIBUTED NOSQL DATABASE

Cassandra



Apache • scalable + fault-tolerant

Apache Cassandra is a highly scalable, distributed NoSQL database that handles **massive data across many servers with no single point of failure**. Its fault-tolerant architecture, high availability, and linear scalability make it ideal for real-time, big-data, and cloud-native systems.

➔ cassandra.apache.org

WHAT IT IS

A distributed, fault-tolerant NoSQL store.

IN AI STACKS

Holds high-volume, real-time application data.

WHY DEVS USE IT

No single point of failure, linear scaling.

BEST FOR

Big-data and cloud-native systems.

↓ UP NEXT • CHANNELS TO FOLLOW

Spotlight next.

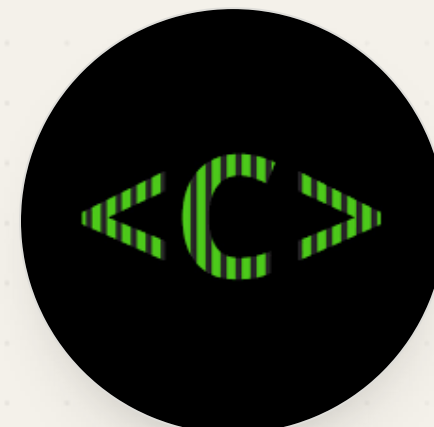
The **high-signal, low-noise** YouTube channels worth a follow to keep your C-stack current.

C

HIGH-SIGNAL, LOW-NOISE

Computerphile

YouTube • computer science



Computerphile is a popular educational channel exploring computer science, AI, cybersecurity, algorithms, and programming through **engaging discussions with leading researchers and educators**. Its accessible explanations and real-world insights make complex technical topics understandable for learners, developers, and enthusiasts alike.

▶ youtube.com/@Computerphile

WHAT THEY POST

Researcher-led explainers on CS and AI.

FIND THEM

YouTube @Computerphile.

WHY FOLLOW

Deep topics made genuinely understandable.

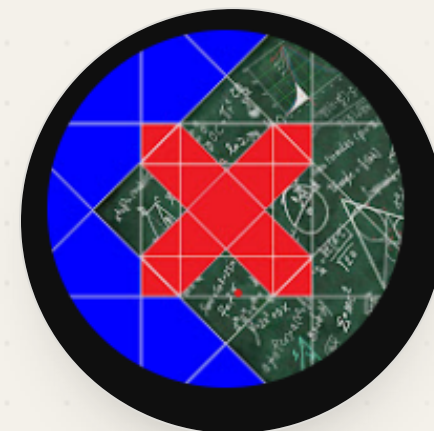
BEST FOR

Building real intuition for fundamentals.

HIGH-SIGNAL, LOW-NOISE

CampusX

India • AI + data science



CampusX is one of India's leading AI and data science education platforms, offering comprehensive courses in **machine learning, deep learning, NLP, MLOps, and data analytics**. Through practical projects, intuitive teaching, and industry-focused content, it helps learners build strong foundations and real-world AI skills.

▶ youtube.com/@campusx-official

WHAT THEY POST

Project-based ML, deep learning, and MLOps.

FIND THEM

YouTube @campusx-official.

WHY FOLLOW

Strong foundations with hands-on practice.

BEST FOR

Learners building real AI skills.

HIGH-SIGNAL, LOW-NOISE

CodeBasics

project-based AI + data



Codebasics is a leading AI and data science education platform that simplifies **machine learning, analytics, SQL, Power BI, and generative AI** through project-based learning. With industry-focused courses, YouTube content, and real-world case studies, it empowers learners to build practical skills and grow their careers.

▶ youtube.com/@codebasics

WHAT THEY POST

Project tutorials on ML, SQL, and Power BI.

FIND THEM

YouTube @codebasics.

WHY FOLLOW

Practical skills you can apply at work.

BEST FOR

Career-focused, hands-on learners.

See you for D.

Save it, share it, and follow along. The 7 C's, five tools that build your stack, and three channels worth following.

01 **Cost Function** foundations

02 **Clustering** learning

03 **CNN** vision

04 **Chain of Thought** prompting

05 **Context Window** llms

06 **Chunking** retrieval

07 **CUDA** infra

+ **Claude Code** tool

+ **Cursor** tool

+ **CrewAI** tool

+ **Cohere** tool

+ **Cassandra** tool

+ **Computerphile** channel

+ **CampusX** channel

+ **CodeBasics** channel



FOLLOW FOR THE FULL A TO Z

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D Next